The University of Sydney

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Portfolio Construction and Optimisation using Market Data and APRA guidelines

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1 Executive Summary

This project investigates how to design an investment strategy that achieves the best balance between risk and return across major asset classes considered by APRA for its performance benchmark. Using historical market data, we estimate the expected returns and risk relationships between these assets and apply them within both static and dynamic portfolio-optimisation frameworks. In the static analysis, we use estimated parameters to construct the efficient frontier and identify optimal portfolios that meet a target return of CPI + 3%. The resulting efficient portfolios highlight the trade-offs between risk and return, forming two benchmark strategies: a defensive and an aggressive portfolio. The dynamic analysis extends this framework to a multi-period setting, where portfolios are rebalanced periodically to account for market drift and transaction costs. Several reallocation policies are tested, including buy-and-hold, annual rebalancing, and conditional rebalancing with risk attribution. Our results show that the annual reallocation with risk attribution delivers the best overall performance, achieving a Sharpe ratio of 0.93 and annual volatility of 7.3%. This strategy balances risk contributions across assets, aligns with the CPI + 3% target return, and offers a practical and efficient approach for long-term investment management.

2 Introduction

Portfolio optimisation is central to investment management, providing a systematic way to allocate wealth across assets to achieve a balance between risk and return. Building on Markowitz's mean–variance theory, investors can identify efficient portfolios that deliver the highest expected return for a given level of risk. This framework remains a core principle in both academic finance and real-world asset allocation.

This project applies these principles to construct optimal portfolios across major growth asset classes, following APRA's benchmark structure. We first estimate each asset's expected return and risk using historical market data, supported by a neural-network-integrated model to improve estimation accuracy. These parameters form the inputs for two optimisation frameworks: static optimisation, which identifies efficient portfolios for a single investment horizon, and dynamic optimisation, which allows periodic rebalancing to reflect changing market conditions and transaction costs. The report is organised as follows. Section 3 presents the parameter-estimation methodology. Section 4 covers static portfolio optimisation, including the efficient frontier and utility-based allocation. Section 5 extends the framework to dynamic strategies incorporating rebalancing and risk attribution. Section 6 presents empirical results, performance evaluation, and key recommendations for achieving the CPI + 3% target return.

3 Parameters Estimation

Traditional models such as Capital Asset Pricing Model (CAPM) and Fama-French Three-Factor Model (FF3) assume a linear factor structure

$$\mathbf{R}_t = \mathbf{B}\mathbf{f}_t + \boldsymbol{\epsilon}_t,$$

where $\mathbf{f}_t$ are common factors and $\mathbf{B}$ are loadings.

Instead, we adopt a nonlinear latent-factor model based on an autoencoder, which learns a data-driven mapping between observed returns and latent representations. Specifically,

$$\mathbf{R}_t = \mathbf{g}_\phi(\mathbf{f}_\theta(\mathbf{R}_t)) + \boldsymbol{\epsilon}_t,$$

where $\mathbf{f}_\theta : \mathbb{R}^N \to \mathbb{R}^K$ is the encoder, mapping returns into K latent factors $\mathbf{z}_t = \mathbf{f}_\theta(\mathbf{R}_t)$; $\mathbf{g}_\phi : \mathbb{R}^K \to \mathbb{R}^N$ is the decoder, reconstructing the input returns from these latent factors; $\boldsymbol{\epsilon}_t$ represents residual reconstruction noise. The autoencoder therefore identifies nonlinear structures in return data without imposing a pre-specified factor model. The autoencoder architecture, training and estimation process are explained in Appendix. Using the latent factor matrix $\mathbf{Z} = \begin{bmatrix}\mathbf{z}_1 & \cdots & \mathbf{z}_T\end{bmatrix}^\top \in \mathbb{R}^{T\times K}$ obtained from rolling-window estimation, we estimate the expected returns and covariance matrix implied by the learned factors.

3.1 Expected Returns

Each asset's return series $\mathbf{R}_i \in \mathbb{R}^T$ is regressed on the latent factors:

$$\mathbf{R}_i = \alpha_i \mathbf{1}_T + \mathbf{Z}\boldsymbol{\beta}_i + \boldsymbol{\epsilon}_i, \quad i = 1, \ldots, N,$$

yielding estimated loadings $\hat{\boldsymbol{\beta}}_i$. Stacking across assets gives the loading matrix $\mathbf{B} = \begin{bmatrix}\hat{\boldsymbol{\beta}}_1^\top & \cdots & \hat{\boldsymbol{\beta}}_N^\top\end{bmatrix} \in \mathbb{R}^{N\times K}$. The average returns $\bar{\mathbf{R}} = \frac{1}{T}\sum_{t=1}^T \mathbf{R}_t$ are the regressed on $\mathbf{B}$:

$$\bar{\mathbf{R}} = \alpha \mathbf{1}_N + \mathbf{B}\boldsymbol{\lambda} + \mathbf{u},$$

producing factor premia $\hat{\boldsymbol{\lambda}}$ and model-implied expected returns

$$\boldsymbol{\mu} = \hat{\alpha}\mathbf{1}_N + \mathbf{B}\hat{\boldsymbol{\lambda}}.$$

This two-stage procedure parallels the Fama-MacBeth regression but uses nonlinear latent factors from the autoencoder.

3.2 Covariance Matrix

To obtain a stable covariance estimate, given the demeaned return matrix $\mathbf{Y} = \mathbf{R}^\top - \boldsymbol{\mu}\mathbf{1}_T^\top$, the sample covariance is defined as

$$\mathbf{S} = \frac{1}{T}\mathbf{Y}\mathbf{Y}^\top \in \mathbb{R}^{N\times N}.$$

Becuse $\mathbf{S}$ can be noisy and ill-conditioned when T is small relative to N, a Principal Component Analysis (PCA) based decomposition is used to extract the dominant factors. Let the eigendecomposition of $\mathbf{S}$ be

$$\mathbf{S} = \mathbf{H}\mathbf{S}_D\mathbf{H}^\top = \sum_{i=1}^N s_i^2 \mathbf{h}_i\mathbf{h}_i^\top,$$

where s_i^2 are the eigenvalues and $\mathbf{h}_i$ the corresponding eigenvectors. The matrix $\mathbf{H} = \begin{bmatrix}s_1\mathbf{h}_1 & \cdots & s_N\mathbf{h}_N\end{bmatrix}$ is sorted so that $s_1^2 \geq s_2^2 \geq \cdots \geq s_N^2$. The top k components capturing the majority of variance (typically until a cumulative threshold $\tau = 0.75$) form $\mathbf{H}_k = \begin{bmatrix}s_1\mathbf{h} & \cdots & s_k\mathbf{h}_k\end{bmatrix}$.

This defines the systematic covariance $\mathbf{S}_{\text{sys}} = \mathbf{H}_k\mathbf{H}_k^\top$. The residual covariance is $\mathbf{N} = \mathbf{S} - \mathbf{S}_{\text{sys}}$, and its diagonal elements represent the specific (idiosyncratic) risks. The diagonal matrix $\mathbf{\Delta} = \text{diag}(\text{diag}(\mathbf{N}))$ preserves only these residual variances, removing cross-asset noise. Finally, the PCA-regularised covariance matrix used for optimisation is

$$\mathbf{\Sigma} = \mathbf{H}_k\mathbf{H}_k^\top + \mathbf{\Delta}.$$

This decomposition ensures that $\mathbf{H}_k\mathbf{H}_k^\top$ captures the low-rank common structure of returns, $\mathbf{\Delta}$ captures uncorrelated idiosyncratic variation, and $\mathbf{\Sigma}$ remains positive semi-definite and well-conditioned.

4 Static Portfolio Optimisation

Consider a portfolio of N risky assets. Let $\boldsymbol{\mu} \in \mathbb{R}^N$ and $\mathbf{\Sigma} \in \mathbb{R}^{N\times N}$ denote, respectively, the expected returns and covariance matrix obtained from our estimation pipeline introduced in Section 3, and $\mathbf{w} = \begin{bmatrix} w_1 & \cdots & w_N \end{bmatrix}^\top$ be the portfolio weight vector with budget constraint $\mathbf{1}^\top\mathbf{w} = 1$. We impose a bound on each weight by the same box, i.e. $L \le w_i \le M$. Assuming that the asset returns follow a multivariate normal distribution, $\mathbf{R} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{\Sigma})$, the portfolio return is given by $R_p = \mathbf{w}^\top\mathbf{R}$ with corresponding expected return $\mu_p = \mathbf{w}^\top\boldsymbol{\mu}$ and variance $\sigma_p^2 = \mathbf{w}^\top\mathbf{\Sigma}\mathbf{w}$. Each feasible weight vector $\mathbf{w}$ thus defines a point (μ_p, σ_p) in the mean-variance space and the portfolio return R_p follows one-dimensional normal distribution, $R_p \sim \mathcal{N}(\mu_p, \sigma_p^2)$.

4.1 Global Minimum-Variance Portfolio

The global minimum-variance (GMV) portfolio minimises risk without a target-return constraint:

$$\min_{\mathbf{w}} \mathbf{w}^\top\mathbf{\Sigma}\mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top\mathbf{w} = 1, \;\; L \le w_i \le M.$$

The solution $\mathbf{w}_{\text{GMV}}$ yields mean $\mu_{\text{GMV}} = \mathbf{w}_{\text{GMV}}^\top\boldsymbol{\mu}$ and variance $\sigma_{\text{GMV}}^2 = \mathbf{w}_{\text{GMV}}^\top\mathbf{\Sigma}\mathbf{w}_{\text{GMV}}$. The efficient frontier (EF) is the set of minimum variances achievable for given returns.

4.2 Minimum Variance Portfolio with Target Return

For a specific target μ_*,

$$\min_{\mathbf{w}} \mathbf{w}^\top\mathbf{\Sigma}\mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top\boldsymbol{\mu} = \mu_*, \;\; \mathbf{1}^\top\mathbf{w} = 1, \;\; L \le w_i \le M.$$

Let $\text{EF}(\mu)$ denote that minimum variance. If $\mu_* \le \mu_{\text{GMV}}$, the GMV portfolio is chosen; otherwise the minimum-variance portfolio (MVP) is the point $(\mu_*, \text{EF}(\mu_*))$:

$$\mathbf{w}_{\text{GMV}} = \begin{cases} \mathbf{w}_{\text{GMV}} & \mu_* \le \mu_{\text{GMV}}, \\ \text{portfolio at } (\mu_*, \text{EF}(\mu_*)) & \text{otherwise.} \end{cases}$$

4.3 Including a fixed weight of Cash

When a fixed proportion $\varphi \in [0,1]$ of total wealth invested in risky assets and the remaining $1-\varphi$ in cash (risk-free asset) with return r_f, the combined portfolio weights become

$$\mathbf{w}^{[\varphi]} = \begin{bmatrix} 1-\varphi \\ \varphi \mathbf{w} \end{bmatrix} \tag{4.1}$$

and the corresponding expected return and standard deviation are transformed as

$$\left(\mu^{[\varphi]}, \sigma^{[\varphi]}\right) = \big((1-\varphi)r_f + \varphi\mu_p,\ \varphi\sigma_p\big), \tag{4.2}$$

representing the coordinates after inclusion of cash. This transformation shifts and scales the EF, GMV portfolio and MVP.

4.4 The Exponential Utility

Under the exponential (CARA) utility $U(x) = -e^{-\gamma x}$ with $\gamma > 0$ and normally distributed $R_p \sim \mathcal{N}(\mu_p, \sigma_p^2)$,

$$\mathbb{E}[U(R_p)] = -\exp\left(-\gamma\mu_p + \frac{1}{2}\gamma^2\sigma_p^2\right), \tag{4.3}$$

so maximising expected utility is equivalent to maximising the certainty equivalent

$$\mathbf{w}^* = \arg\max_{\mathbf{w}}\left(\mu_p - \frac{1}{2}\gamma\sigma_p^2\right) = \arg\max_{\mathbf{w}}\left(\mathbf{w}^\top\boldsymbol{\mu} - \frac{1}{2}\gamma\mathbf{w}^\top\boldsymbol{\Sigma}\mathbf{w}\right).$$

4.5 Efficient Frontier and utility-based solution

The EF for risky assets is computed numerically using Sequential Least Squares Quadratic Programming (SLSQP) via `scipy.optimize.minimize`, given $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. Applying the linear transform (4.1) and (4.2) extends the EF to portfolios that include a fixed cash share. Thus, the GMV point maps to

$$\left(\mu_{\text{GMV}}^{[\varphi]}, \sigma_{\text{GMV}}^{[\varphi]}\right) = \big((1-\varphi)r_f + \varphi\mu_{\text{GMV}},\ \varphi\sigma_{\text{GMV}}\big),$$

showing that introducing cash scales both expected return and volatility proportionally to φ.

With CARA utility, using (4.3), the expected utility of a portfolio that includes cash is

$$\mathbb{E}[U(x)] = -\exp\left[-\gamma\big((1-\varphi)r_f + \varphi\mathbf{w}^\top\boldsymbol{\mu}\big) + \frac{1}{2}\gamma^2\varphi^2\mathbf{w}^\top\boldsymbol{\Sigma}\mathbf{w}\right],$$

where $x = (1-\varphi)r_f + \varphi\mathbf{w}^\top\mathbf{R} \sim \mathcal{N}\big((1-\varphi)r_f + \mathbf{w}^\top\boldsymbol{\mu},\ \varphi^2\mathbf{w}^\top\boldsymbol{\Sigma}\mathbf{w}\big)$ is the total return including cash, Maximising expected utility is therefore equivalent to

$$\min_{\mathbf{w},\varphi}\mathbb{E}[U(x)] \quad \text{s.t.} \quad \mathbf{1}^\top\mathbf{w} = 1,\ \ L \le w_i \le M.$$

Since we consider bounds to be active, we solve this problem numerically.

4.6 Sharpe Ratio

To assess the risk-adjusted performance of a portfolio, we compute the Sharpe Ratio, which measures the excess return earned per unit of risk taken by the portfolio.

$$\mathrm{SR}(\mathbf{w}) = \frac{\mu_p - r_f}{\sigma_p} = \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}}$$

5 Dynamic Portfolio Optimisation

Let $\{\mathbf{V}_t\}_{t=0}^N$ denote the n-dimensional excess-return vectors of risky assets and $\{\mathbf{w}_t\}_{t=1}^N$ the portfolio weights chosen at the beginning of each period, adapted to the information filtration $\{\mathcal{F}_t\}$. A fixed fraction $\varphi \in [0,1]$ of total wealth is invested in the risky assets each period, with the remaining $1-\varphi$ in a risk-free asset yielding r_f. We consider exponential utility (CARA) $U(x) = -e^{-\gamma x}$ with $\gamma > 0$. Throughout, expected returns and covariances are held fixed $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. Define the within-period gross wealth factor

$$X_t := 1 + (1-\varphi) r_f + \varphi \mathbf{w}_t^\top \mathbf{V}_t,$$

and the multiplicative wealth process

$$Z_t := \prod_{k=1}^{t} X_k, \quad Z_0 := 1.$$

The terminal objective is to choose the control sequence to maximise expected utility of terminal wealth:

$$\max_{\{\mathbf{w}_t\}_{t=1}^N} \mathbb{E}[U(Z_N - 1)], \quad \mathbf{w}_t \in \mathcal{W}_t, \;\; t = 1, \ldots, N.$$

5.1 Bellman recursion

Dynamic programming yields a backward recursion. For $t = N, \ldots, 1$, define the value function at current wealth level Z_t^{fix} by

$$V_t(Z_t^{\mathrm{fix}}) := \sup_{\mathbf{w}_t, \ldots, \mathbf{w}_N} \mathbb{E}\left[U\left(Z_t^{\mathrm{fix}} \prod_{k=t}^{N} X_k - 1 \right) \,\middle|\, \mathcal{F}_{t-1} \right], \;\; V_{N+1}(Z_{N+1}^{\mathrm{fix}}) := U(Z_{\mathrm{N+1}}^{\mathrm{fix}} - 1).$$

Conditioning on $\mathcal{F}_{t-1}$, the only random variable at stage t is $\mathbf{V}_t$. Thus the Bellmand recursion reduce s to

$$V_t(Z_t^{\mathrm{fix}}) = \sup_{\mathbf{w}_t \in \mathcal{W}_t} \mathbb{E}\big[V_{t+1}(Z_t^{\mathrm{fix}} X_t) \mid \mathcal{F}_{t-1} \big]. \tag{5.1}$$

Under CARA utility with normal returns, the continuation value is exponential-affine. Assume

$$V_{t+1}(Z_t^{\mathrm{fix}}) = -K_{t+1} e^{-\gamma Z_t^{\mathrm{fix}}}, \quad K_{t+1} > 0.$$

Substituting this form into the Bellman step gives an objective proportional to

$$\mathbb{E}\Big[e^{-\gamma Z_{t-1}^{\mathrm{fix}}(X_t - 1)} \mid \mathcal{F}_{t-1} \Big].$$

If $\mathbf{V}_t \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then

$$X_t - 1 = (1-\varphi)r_f + \varphi \mathbf{w}_t^\top \mathbf{V}_t \sim \mathcal{N}\big((1-\varphi)r_f + \varphi \mathbf{w}_t^\top \boldsymbol{\mu},\ \varphi^2 \mathbf{w}_t^\top \boldsymbol{\Sigma} \mathbf{w}_t\big).$$

Using the normal moment generating function,

$$\mathbb{E}\Big[\exp\big(-\gamma Z_{t-1}^{\text{fix}}(X_t - 1)\big)\Big] \propto \exp\Big[-\gamma Z_{t-1}^{\text{fix}}((1-\varphi)r_f + \varphi \mathbf{w}_t^\top \boldsymbol{\mu}) + \frac{1}{2}(\gamma Z_{t-1}^{\text{fix}}\varphi)^2 \mathbf{w}_t^\top \boldsymbol{\Sigma} \mathbf{w}_t\Big].$$

Since $\exp(\cdot)$ is increasing, maximising the value function is equivalent, for each t, to the quadratic program

$$\max_{\mathbf{w}_t \in \mathcal{W}_t} \gamma Z_{t-1}^{\text{fix}} \varphi \boldsymbol{\mu}^\top \mathbf{w}_t - \frac{1}{2}(\gamma Z_{t-1}^{\text{fix}}\varphi)^2 \mathbf{w}_t^\top \boldsymbol{\Sigma} \mathbf{w}_t.$$

Hence, the optimal decision at time t is a local mean-variance trade-off in growth space with effective risk aversion scaled by current wealth Z_{t-1}^{fix}.

5.2 Target drift

Let the initial allocation including cash be $\mathbf{w}^{[\varphi]} = \begin{bmatrix} 1-\varphi & \varphi \mathbf{w} \end{bmatrix}^\top$ where $\mathbf{w}$ is the risky-assets allocation applied to $\mathbf{V}_1, \ldots, \mathbf{V}_t$. With no rebalancing, compounding induces drift in the composition. After t periods the drifted weights are

$$\text{TD}_t(\mathbf{w}^{[\varphi]}) = \frac{1}{(1-\varphi)\prod_{k=1}^t (1+r_f) + \varphi \prod_{k=1}^t (1 + \mathbf{w}^\top \mathbf{V}_k)} \begin{bmatrix} (1-\varphi)\prod_{k=1}^t (1+r_f) \\ \varphi \prod_{k=1}^t (1 + \mathbf{w}^\top \mathbf{V}_k) \end{bmatrix},$$

which captures the cumulative deviation from the target due to unequal growth between defensive and risky assets.

5.3 Reallocation

Let the desired dixed target be $\hat{\mathbf{w}}^{[\varphi]} = \begin{bmatrix} 1-\varphi & \varphi \hat{\mathbf{w}} \end{bmatrix}^\top$ and let $\text{TD}_1(\cdot)$ denote one-period drift. The trade needed to fully store the fixed target after one period is $\hat{\mathbf{w}}^{[\varphi]} - \text{TD}_1(\hat{\mathbf{w}}^{[\varphi]})$.

If instead the aim is to track the stochastic optimal weights $\mathbf{w}_t^{[\varphi]*}$ obtained from the quadratic program above, the one-period adjustment becomes $\mathbf{w}_t^{[\varphi]} - \text{TD}_1(\hat{\mathbf{w}}_{t-1}^{[\varphi]*})$.

For rebalancing every k periods, the cumulative drift corrected at time $t+k$ is

$$\mathbf{w}_{t+k}^{[\varphi]*} - \text{TD}_k(\hat{\mathbf{w}}_t^{[\varphi]*}).$$

5.4 Turnover rate

The rebalancing intensity is summarised by the turnover rate

$$\text{TR} := \frac{1}{2} \sum_i \Big| \hat{w}_i^{[\varphi]} - [\text{TD}_1(\mathbf{w}^{[\varphi]})]_i \Big|,$$

interpretable as the fraction of wealth traded to restore the target each period.

5.5 Reallocation fee

With proportional transaction cost $c > 0$ per unit traded, the post-fee wealth factor in period t is $\widetilde{X}_t = X_t(1 - c\,\mathrm{TR}_t)$ and the wealth recursion becomes

$$Z_{t+1} = Z_t \widetilde{X}_t = Z_t\big[(1-\varphi)(1+r_f) + \varphi(1 + \mathbf{w}_t^\top \mathbf{V}_t)\big](1 - c\,\mathrm{TR}_t).$$

Expected utility is evaluated on $\{Z_t\}$ under this adjustment. Trading frictions therefore penalise high-turnover policies and can shift the dynamically rebalancing frequency toward more conservative schedules.

6 Computational Results

6.1 Parameters Estimation

The latent-factor autoencoder was trained over a rolling 10-year window to extract nonlinear market factors and estimate the implied expected returns and PCA was applied to estimate the covariance matrix. Model diagnostics and validation plots are shown below.

$$\boldsymbol{\Sigma} = 10^{-4} \times \begin{bmatrix} 17.71 & 13.03 & 13.68 & 14.70 & 6.50 & 4.07 \\ 13.03 & 20.30 & 16.56 & 11.79 & 7.15 & 4.57 \\ 13.68 & 16.56 & 31.95 & 9.96 & 5.83 & 5.51 \\ 14.70 & 11.79 & 9.96 & 37.89 & 8.72 & 3.73 \\ 6.50 & 7.15 & 5.83 & 8.72 & 10.34 & 2.30 \\ 4.07 & 4.57 & 5.51 & 3.73 & 2.30 & 3.15 \end{bmatrix}, \qquad \boldsymbol{\mu} = \begin{bmatrix} 0.0073 \\ 0.0054 \\ 0.0096 \\ 0.0057 \\ 0.0062 \\ 0.0046 \end{bmatrix}.$$

6.1.1 Normality and Factor Diagnostics

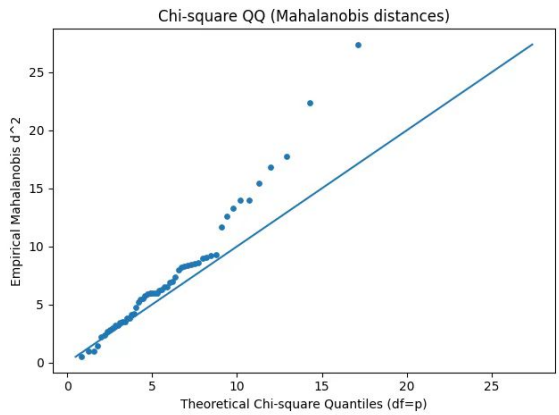
To verify the multivariate normality assumption underlying the mean-variance framework, we compute the Mahalanobis distances of latent returns and compare them to a theoretical chi-square distribution with $p = N$ degrees of freedom. The resulting Chi-square Q-Q plot (Figure 1(a)) shows that most observations align closely with the 45-degree reference line, confirming approximate elliptical symmetry and supporting the validity of the Normal assumption for latent-factor modelling.

6.1.2 Expected Return Validation

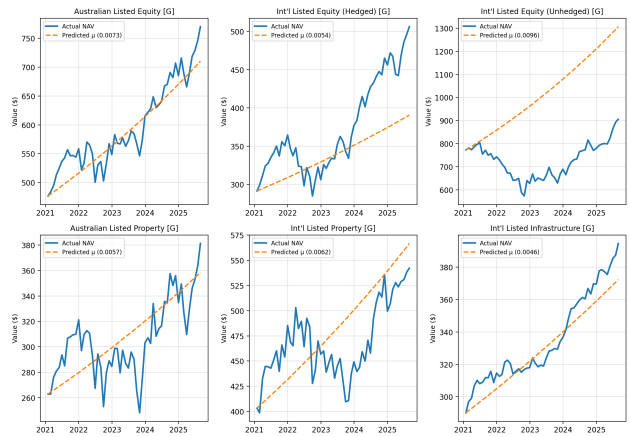
Figure 1(b) compares the actual NAV trajectories of each growth asset with the predicted mean return $\boldsymbol{\mu}$ estimated by the latent factors autoencoder model. The orange dashed lines represent the expected growth paths implies by $\boldsymbol{\mu}$, while blue lines show realised NAVs.

Across six growth asset classes, the latent-factor model captures the general reasonable expected return estimates consistent with market trends between 2021 and 2025. Minor deviations arise

during periods of market stress (e.g. 2022-2023 volatility), which are expected since the autoencoder estimates parameters from historical information only. These results confirm that, the autoencoder successfully identifies stable latent factors consistent with underlying market structure. Estimated expected returns track long-term growth trends across assets, the estimated covariance remains well-conditioned for use in mean-variance optimisation. Together, these diagnostics validate the statistical soundness of the autoencoder-based parameter estimation before proceeding to static and dynamic portfolio optimisation in next few subsections.



(a) Chi-square Q–Q plot of Mahalanobis distances for latent-factor returns.



(b) Comparison between realised NAVs and model-implied expected returns ($\hat{\mu}$).

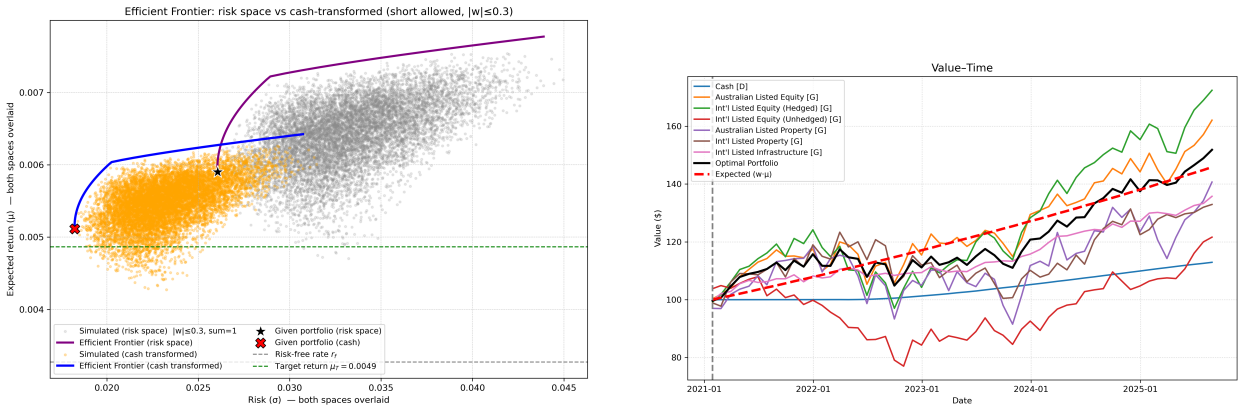
Figure 1: Parameter estimation diagnostics: model normality and expected-return validation.

6.2 Static Optimisation

This section reports the results for the static problem using the estimated growth-asset parameters $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, monthly risk-free rate $r_f \approx 0.0033$, and per-asset bounds $(-0.1, 0.3)$. We impose budget and box constraints on risky weights. Cash is added ex post via the transformation (4.2) with baseline cash share $1 - \varphi = 0.3$ unless otherwise noted.

6.2.1 Efficient Frontier and Target Mapping

Figure 2(a) overlays two spaces. The grey cloud shows 10,000 feasible risky portfolios under $-0.1 \leq w_i \leq 0.3$, $\mathbf{1}^\top \mathbf{w} = 1$. The purple curve is the corresponding EF obtained by sweeping target μ and solving the minimum-variance SLSQP with the same constraints. The orange cloud is the cash-mapped version for $\varphi = 0.7$; the blue curve is the implied cash-space EF. The dashed lines mark r_f (grey) and our policy target $\mu_T \approx 0.0049$ per month (green), which stems from CPI + 3% per asset. The black star locates the optimised risky mix at the target in risky space; its red cross is the image after adding 30% cash. This figure shows that the constrints noticeably bend the EF relative to the unconstrained parabela; also, introducing cash shifts up intercepts by $(1 - \varphi)r_f$ and scales risk linearly by φ, so that moving along φ traces a straight ray through the risky point.



(a) Efficient frontiers.

(b) NAV through time.

Figure 2: Static portfolio-optimisation results: efficient frontier and corresponding NAV performance.

6.2.2 Out-of-sample NAV and realised performance

In Figure 2(b), we start at 100 on the cut date (2021-01, vertical dashed line) and plot post-cut cumulative values of each sector, the cash line, and the optimised cash-augmented portfolio. The black line is the portfolio Net Asset Value (NAV); the red dashed line is the deterministic curve using $\mathbf{w}^\top \boldsymbol{\mu}$ (i.e., the expected monthly growth of the static mix including cash). The optimised portfolio compounds smoothly and, over the evaluation window, attains an annualised Sharpe of about 0.51, with annualised return and volatility consistent with the mean/variance predicted by $(\mu^{[\varphi]}, \sigma^{[\varphi]})$. As expected, cash contributes a steady slope while equities and listed property/infrastructure drive cyclical deviations around the red expectation line.

6.2.3 Defensive (30% growth) versus Aggressive (90% growth)

In Figure 3, grey points are feasible risky portfolios under $-0.1 \le w_i \le 0.3$ and $\mathbf{1}^\top \mathbf{w} = 1$; orange points are the cash-mapped versions for a baseline φ. Blue dashed lines are iso-Sharpe lines,

$$\mathrm{SR} = \frac{\mu^{[\varphi]} - r_f}{\sigma^{[\varphi]}} = \frac{\varphi(\mu_p - r_f)}{\varphi \sigma_p} = \frac{\mu_p - r_f}{\sigma_p},$$

so they are straight lines through the intercept r_f with common slope and therefore do not depend on φ. This visualises why defensive ($\varphi = 0.3$) and aggressive ($\varphi = 0.9$) portfolios have the same Sharpe as the underlying risk bundle.

Green solid curves are iso-CARA indifference curves for exponential utility

$$\mathbb{E}[U] = -\exp\left[-\gamma((1-\varphi)r_f + \varphi\mu_p) + \frac{1}{2}(\gamma\varphi\sigma_p)^2\right].$$

Holding utility constant yields

$$(1-\varphi)r_f + \varphi\mu - \frac{1}{2}\gamma\varphi^2\sigma^2 = \text{constant} \implies \mu = r_f + \frac{\text{constant}}{\varphi} + \frac{1}{2}\gamma\varphi\sigma^2,$$

which are upward-opening parabolas in the (σ, μ) plane with curvature proportional to φ. Higher φ makes the indifference curve upward more steeply. For the same (σ, μ), an aggressive portfolio attains higher CARA utility than a defensive portfolio whenever the risky bundle has positive excess return.

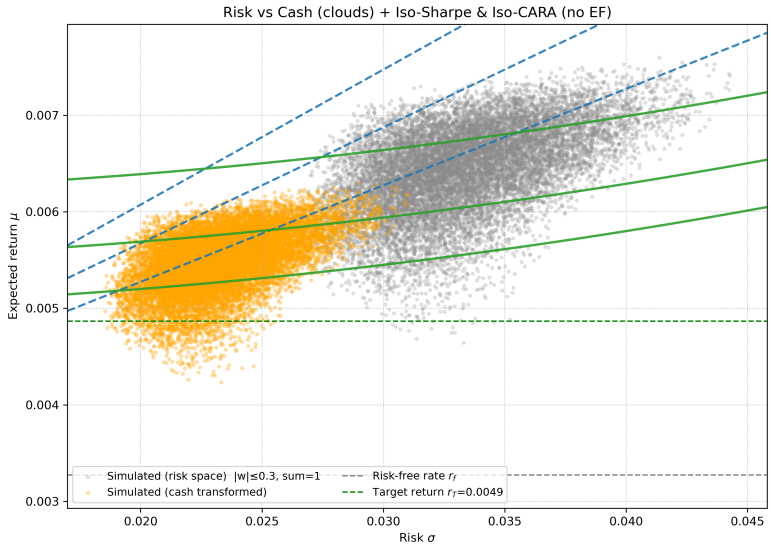


Figure 3: Risk–return simulation with iso-Sharpe and iso-CARA contours

6.3 Dynamic Optimisation

Dynamic optimisation extends the static framework by allowing portfolio weights to adjust over time in response to market movements and transaction costs. Our aim is to identify a reallocation rule that maintains efficiency while limiting turnover.

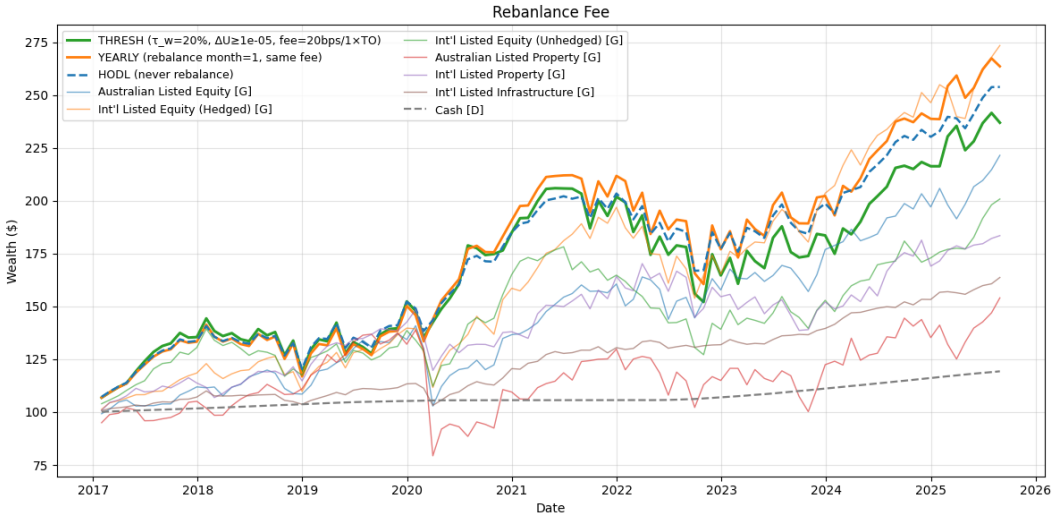


Figure 4: Portfolio wealth after applying a 20 bps transaction fee per unit turnover. Threshold rebalancing remains efficient while reducing cost drag relative to full rebalancing.

Figure 4 shows the impact of rebalancing frequency and cost on portfolio wealth. The threshold and yearly reallocation strategies achieve performance comparable to the full-rebalancing benchmark but with significantly lower lower trading intensity. When a 20 bps transaction fee per 1 turnover is applied, the threshold rule maintains robust growth while controlling cost drag, confirming that disciplined reallocation can enhance long-term efficiency.

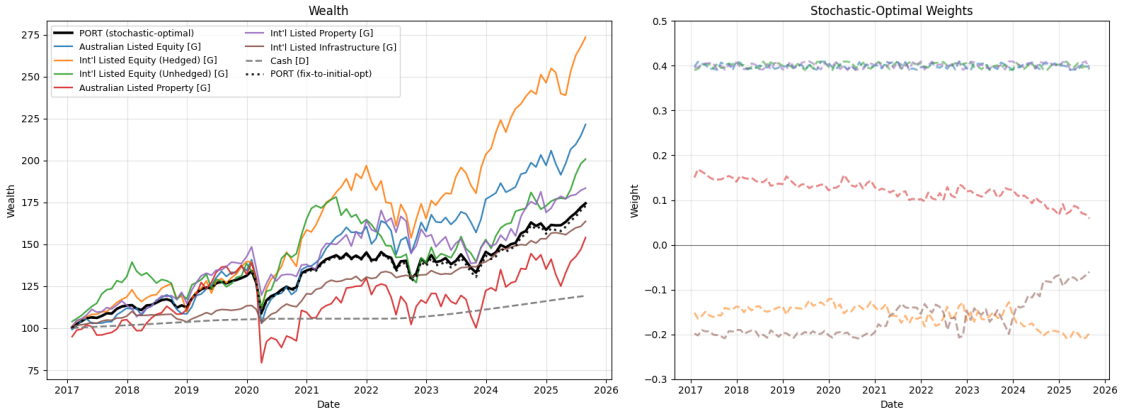


Figure 5: (Left) Wealth trajectories for stochastic-optimal and fixed portfolios. (Right) Time evolution of optimal portfolio weights. The stochastic-optimal solution gradually adjusts allocations while maintaining diversification.

Figure 5 presents the stochastic-optimal solution derived from the dynamic-utility model. The left panel compares wealth trajectories of the stochastic-optimal portfolio and its fixed-weight counterpart, while the right panel tracks the evolution of optimal weights. The stochastic-optimal strategy adjusts allocations gradually, reducing exposure to volatile assets and increasing weights in more stable sectors as time progresses, illustrating the behaviour predicted by the CARA-Normal framework.

Together, these results demonstrate that periodic, risk-aware reallocation yields superior risk-adjusted outcomes compared with passive or overly frequent trading, supporting the adoption of annual rebalancing with risk-attribution constraints as the preferred practical policy.

6.4 Comparison

	Static	Yearly REBA RA	Yearly REBA	Hold
Sharpe Ratio	0.90	0.93	0.98	1.0
Annual Return	8.96%	6.5%	10%	9%
Annual Volatility	9.16%	7.3%	16%	13%

Table 1. Sharpe ratio, annualised return and volatility comparison between strategies.

Table 1 compares the performance metrics across the static and dynamic allocation strategies. The key indicators are the annualised Sharpe ratio, return, and volatility, all computed from the simulated portfolio paths. The static portfolio achieves an annual return of 8.96% with volatility of

9.16%, resulting in a Sharpe ratio of 0.90. This represents a well-balanced allocation that remains efficient without any rebalancing.

The yearly reallocation with risk attribution (Yearly REBA RA) delivers a similar risk-adjusted performance with a slightly lower annual return (6.5%) but the lowest volatility (7.3%). By distributing risk more evenly across assets, this approach achieves a Sharpe ratio of 0.93, which is higher than the static case, indicating better efficiency relative to risk.

In contrast, the plain yearly reallocation (Yearly REBA) achieves a higher return (10%) but at the cost of significantly higher volatility (16%), leading to a lower risk-adjusted outcome.

Finally, the buy-and-hold (Hold) strategy produces a 9% return with 13% volatility, performing reasonably but with higher risk exposure over time. Overall, the yearly reallocation with risk attribution offers the most balanced trade-off between stability and performance. It reduces concentration risk, maintains exposure to growth assets, and achieves a Sharpe ratio comparable to the higher-return but more volatile strategies. This confirms its suitability as the recommended approach for long-term portfolio management aligned with the CPI + 3% target.

7 Conclusion

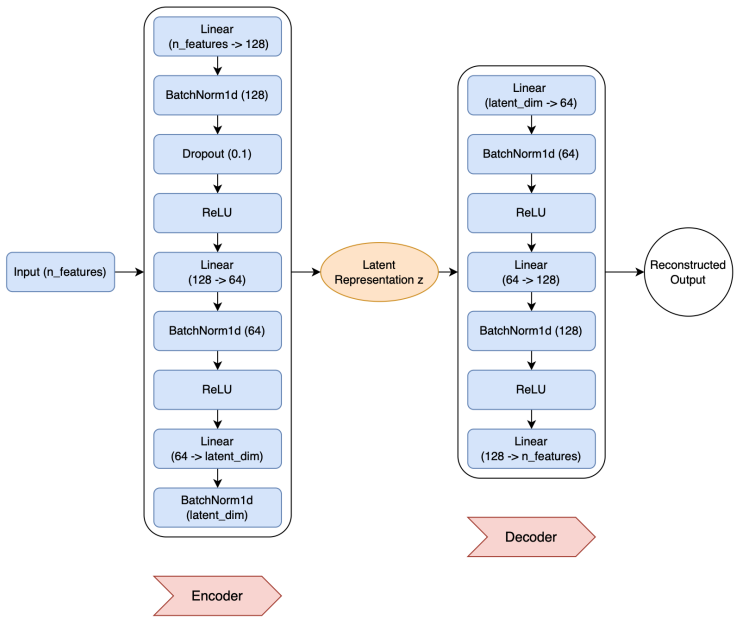
This report applied both static and dynamic portfolio optimisation frameworks to evaluate long-term investment strategies across key growth asset classes. The static analysis constructed the efficient frontier and identified optimal portfolios targeting a return of CPI + 3%, providing a benckmark for risk-return trade-offs. The dynamic analysis extended this framework by allowing periodic rebalancing under realistic market conditions and transaction costs.

Among all tested strategies, the yearly reallocation with risk-attribution constraints achieved the most stable and efficient performance. It produced a Sharpe ratio of 0.93 with annual volatility of 7.3%, outperforming the static portfolio in risk-adjusted terms while maintaining diversification across assets. Although its return is slightly lower than the unconstrained reallocation case, its improved stability and lower turnover make it a practical and robust strategy for achieving the long-term investment objective of CPI + 3%.

8 Appendix: Deep Learning-Based Latent-Factor Autoencoder

Network Architecture

The autoencoder consists of two multilayer perceptrons—an encoder $\mathbf{f}_\theta$ and a decoder $\mathbf{g}_\phi$. Each layer performs a sequence of linear transformations and nonlinear activations designed to compress and then reconstruct the input vector $\mathbf{R}_t$.



For input $\mathbf{x} = \mathbf{R}_t \in \mathbb{R}^N$, the encoder computes successive hidden representations

$$
\begin{aligned}
\mathbf{h}^{(1)} &= \mathrm{ReLU}(\mathrm{BN}^{(1)}(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)})),\\
\widetilde{\mathbf{h}}^{(1)} &= \mathbf{D}^{(1)}(\mathbf{h}^{(1)}),\\
\mathbf{h}^{(2)} &= \mathrm{ReLU}(\mathrm{BN}^{(2)}(\mathbf{W}^{(2)}\widetilde{\mathbf{h}}^{(1)} + \mathbf{b}^{(2)})),\\
\mathbf{z} &= \mathrm{BN}^{(3)}(\mathbf{W}^{(3)}\mathbf{h}^{(2)} + \mathbf{b}^{(3)})
\end{aligned}
$$

where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are the weight matrix and bias vector of layer l; $\mathrm{ReLU}(u) = \max(0, u)$ is the rectified linear unit activation, introducing nonlinearity; $\mathbf{D}^{(1)}(\cdot)$ denotes the dropout operator; $\mathrm{BN}^{(l)}(\cdot)$ denotes Batch Normalisation, defined below; and $\mathbf{z} \in \mathbb{R}^K$ is the latent representation (the output of the encoder). The layer dimensions are $N \xrightarrow{\mathbf{W}^{(1)}} 128 \xrightarrow{\mathbf{W}^{(2)}} 64 \xrightarrow{\mathbf{W}^{(3)}} K$.

The decoder reconstructs the input from $\mathbf{z}$ through symmetric transformations:

$$
\begin{aligned}
\mathbf{h}^{(3)} &= \mathrm{ReLU}(\mathrm{BN}^{(4)}(\mathbf{W}^{(4)}\mathbf{z} + \mathbf{b}^{(4)})),\\
\mathbf{h}^{(4)} &= \mathrm{ReLU}(\mathrm{BN}^{(5)}(\mathbf{W}^{(5)}\mathbf{h}^{(3)} + \mathbf{b}^{(5)})),\\
\widehat{\mathbf{R}}_t &= \mathbf{W}^{(6)}\mathbf{h}^{(4)} + \mathbf{b}^{(6)},
\end{aligned}
$$

yielding the reconstructed returns $\widehat{\mathbf{R}}_t \in \mathbb{R}^N$. The decoder dimensions mirror the encoder: $K \xrightarrow{\mathbf{W}^{(4)}} 64 \xrightarrow{\mathbf{W}^{(5)}} 128 \xrightarrow{\mathbf{W}^{(6)}} N$.

For any hidden activation vector $\mathbf{u} = \begin{bmatrix} u_1 & \cdots u_d \end{bmatrix}^\top$ in a given layer and mini-batch $B = \{u^{(1)}, \ldots, u^{(m)}\}$, Batch Normalisation standardises each component:

$$\mathrm{BN}(u^{(j)}) = \gamma \hat{u}^{(j)} + \beta, \quad \hat{u}^{(j)} = \frac{u^{(j)} - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad \mu_B = \frac{1}{m}\sum_{j=1}^{m} u^{(j)}, \quad \sigma_B^2 = \frac{1}{m}\sum_{j=1}^{m}(u^{(j)} - \mu_B)^2.$$

where $\epsilon > 0$ ensures numerical stability and γ, β are learnable scaling and shifting parameters. This operation keeps the distribution of activations stable across layers, improving training speed and convergence. To prevent overfitting, dropout randomly removes a proportion p of neurons during training. Mathematically, $\mathbf{D}^{(1)}(\mathbf{h}^{(1)}) = \mathbf{m} \odot \mathbf{h}^{(1)}$, where $\mathbf{m}$ is a vector of binary masks ($m_i \sim \mathrm{Bernoulli}(1-p)$) and $\odot$ denotes element-wise multiplication. At inference time, all neurons are used but scaled by $1-p$ to maintain the expected activation level. In this model, we set $p = 0.1$.

The entire autoencoder defines a nonlinear transformation $\widehat{\mathbf{R}}_t = \mathbf{g}_\phi(\mathbf{f}_\theta(\mathbf{R}_t))$, where $\mathbf{f}_\theta$ and $\mathbf{g}_\phi$ comprise the parameters

$$\theta = \{\mathbf{W}^{(1)}, \mathbf{W}^{(2)}, \mathbf{W}^{(3)}, \mathbf{b}^{(1)}, \mathbf{b}^{(2)}, \mathbf{b}^{(3)}\} \quad \text{and} \quad \phi = \{\mathbf{W}^{(4)}, \mathbf{W}^{(5)}, \mathbf{W}^{(6)}, \mathbf{b}^{(4)}, \mathbf{b}^{(5)}, \mathbf{b}^{(6)}\}.$$

Training Objective

Training adjusts these parameters to minimise the reconstruction and regularisation losses described below. The autoencoder parameters (θ, ϕ) are obtained by minimising a joint objective that balances reconstruction accuracy and structural regularisation:

$$\mathcal{L}(\theta, \phi) = \frac{1}{T}\sum_{t=1}^{T} \|\mathbf{R}_t - \widehat{\mathbf{R}}_t\|_2^2 + \lambda \frac{1}{T^2}\|\mathbf{Z}\mathbf{Z}^\top - \mathbf{R}\mathbf{R}^\top\|_F^2,$$

where $\widehat{\mathbf{R}}_t = \mathbf{g}_\phi(\mathbf{f}_\theta(\mathbf{R}_t))$, $\mathbf{R} = \begin{bmatrix} \mathbf{R}_1 & \cdots & \mathbf{R}_T \end{bmatrix}^\top$, $\mathbf{Z} = \begin{bmatrix} \mathbf{z}_1 & \cdots & \mathbf{z}_T \end{bmatrix}^\top$, $\|\cdot\|_F$ is the Frobenius norm, and $\lambda > 0$ controls regularisation strength. The second term encourages the latent factors $\mathbf{Z}$ to reproduce the correlation structure of observed returns.

Rolling-Window Estimation

To account for time-varying market structure, the autoencoder is re-trained on overlapping windows of fixed length W months (typically $W = 120$). Let the training data be $\mathbf{R}_{\text{train}} = \begin{bmatrix} \mathbf{R}_1 & \cdots & \mathbf{R}_T \end{bmatrix}^\top \in \mathbb{R}^{T \times N}$. Define window endpoints $t_j = W + (j-1)P$ with forward step P (e.g $P = 21$ months), and let $\mathbb{W}_j = \{t_j - W + 1, \ldots, t_j\}$. For each window j, scale returns within the window as

$$\widetilde{\mathbf{R}}^{(j)} = \mathrm{diag}(\hat{\boldsymbol{\sigma}}^{(j)})^{-1}(\mathbf{R}^{(j)} - \mathbf{1}_W(\boldsymbol{\mu}^{(j)})^\top),$$

where $\boldsymbol{\mu}^{(j)}$ and $\hat{\boldsymbol{\sigma}}^{(j)}$ are sample means and standard deviations respectively. Then, train an independent autoencoder on $\widetilde{\mathbf{R}}^{(j)}$ to obtain window-specific latent factors $\mathbf{z}_t = \mathbf{f}_{\theta^{(j)}}(\widetilde{\mathbf{R}}_t)$, $t \in \mathbb{W}_j$ by

minimising the loss

$$\mathcal{L}^{(j)} = \frac{1}{W} \sum_{t \in \mathbb{W}_j} \|\widetilde{\mathbf{R}}_t - \widehat{\mathbf{R}}_t\|^2 + \lambda \frac{1}{W^2} \|\mathbf{Z}\mathbf{Z}^\top - \widetilde{\mathbf{R}}\widetilde{\mathbf{R}}^\top\|_F^2.$$

Finally, combine all window outputs into $\mathbf{Z} = \begin{bmatrix} \mathbf{Z}^{(1)} & \cdots & \mathbf{Z}^{(J)} \end{bmatrix}^\top \in \mathbb{R}^{T \times K}$, forming the time-series of latent market factors. This procedure adapts the latent-factor representation to changing market conditions while maintaining temporal consistency for subsequent parameter estimation.

9 References

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